

Signal Flow for Preset Objects:

Preset 1 - Bells

- Our code takes a triangle wave and passes it through the FM equation, using 4 as the modulation ratio and 6 as the modulation index. That signal is then passed through an ADSR envelope of (0, 20, 0.2, 80). This gives us the distinct ringing tone of a bell.

Preset 2 - Buzzy Tremolo

- Our code takes a square wave and passes it through amplitude modulation, using 0.05 as the modulation ratio and 100 as the modulation index. This signal is then passed through an ADSR envelope of (50, 10, 0.8, 10). This creates a buzzy tone that oscillates between pitches to create a tremolo.

Three WAV File Descriptions:

MUSI2526Bells.wav:

- For this audio file, we used Preset 1 - Bells. Nothing else was typed into the CLI (besides file name) because the preset covers all creating of audio, no user interference required.

MUSI2526FM1.wav:

- In creating this file, we made a custom sound using a saw wave. It uses FM modulation synthesis, with a modulation rate of 6 and a harmonicity of 3.5 for slight inharmonic effects. The ADSR was set to (30, 50, 0.7, 20). There is also a third-order lowpass filter with a cutoff of 800Hz. Additionally, there is a reverb designed with preset convolution and a dry/wet mix of 0.5. Finally, there is a single delay time of 350ms with a 0.5 dry/wet mix.

MUSI2526AM1.wav:

- To build this file, we made a custom sound using a square wave. It uses AM synthesis, with a modulation rate of 7 and a harmonicity of 2 for denser, tighter sidebands. The ADSR was set to (60, 30, 0.7, 10). There is also a second-order highpass filter with a cutoff of 200Hz. Finally, there is a single delay time of 250ms with a 0.8 dry/wet mix.